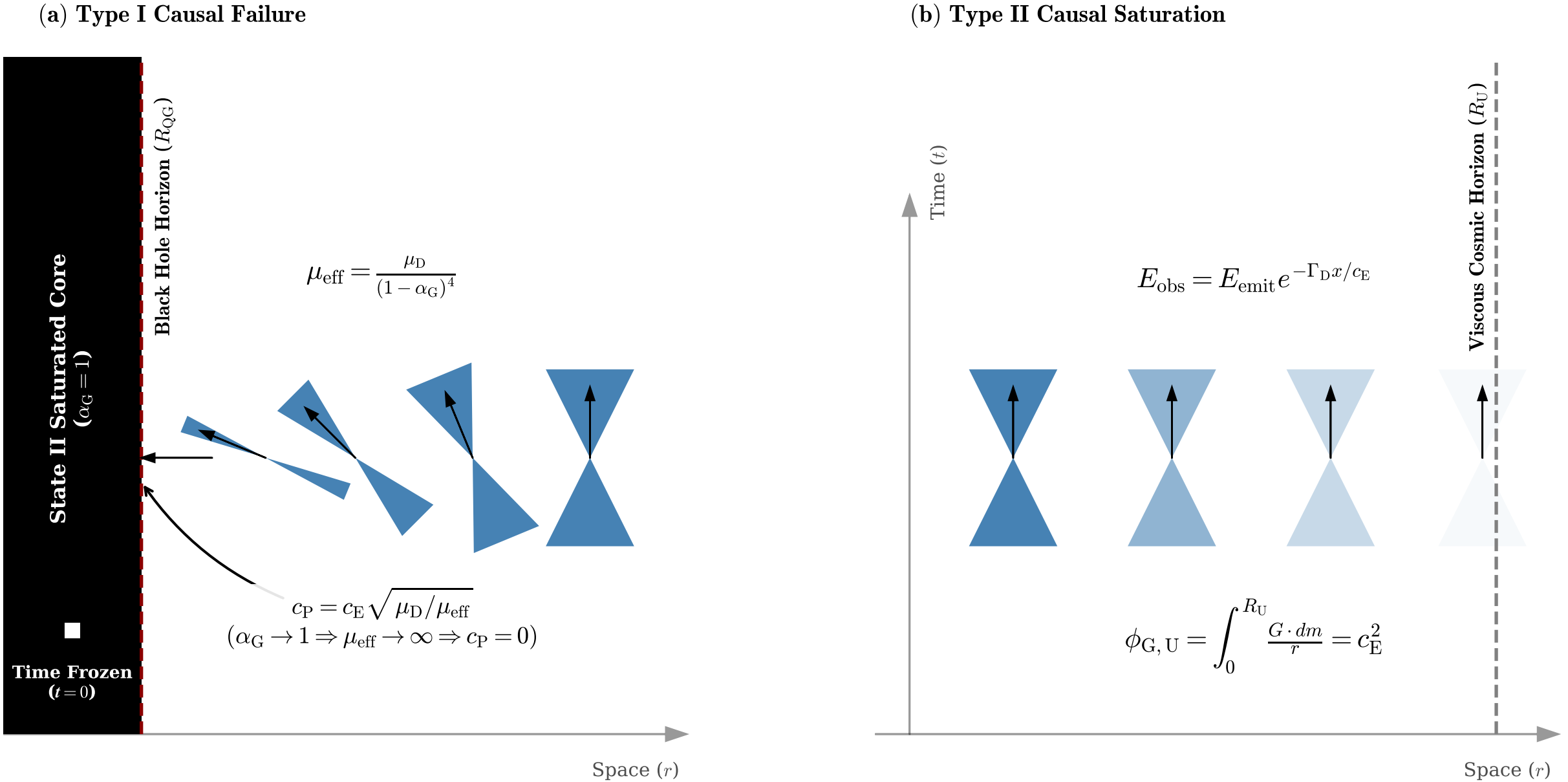

(a) Type I Causal Failure
State II Saturated Core
($\alpha_G = 1$)
Black Hole Horizon (R_{QG})
$\mu_{eff} = \frac{\mu_D}{(1-\alpha_G)^4}$
$c_P = c_E \sqrt{\mu_D / \mu_{eff}}$
($\alpha_G \to 1 \Rightarrow \mu_{eff} \to \infty \Rightarrow c_P = 0$)
Time Frozen
($t = 0$)
Space (r)
(b) Type II Causal Saturation
Time (t)
Viscous Cosmic Horizon (R_U)
$E_{obs} = E_{emit} e^{-\Gamma_D x / c_E}$
$\phi_{G,U} = \int_0^{R_U} \frac{G \cdot dm}{r} = c_E^2$
Space (r)